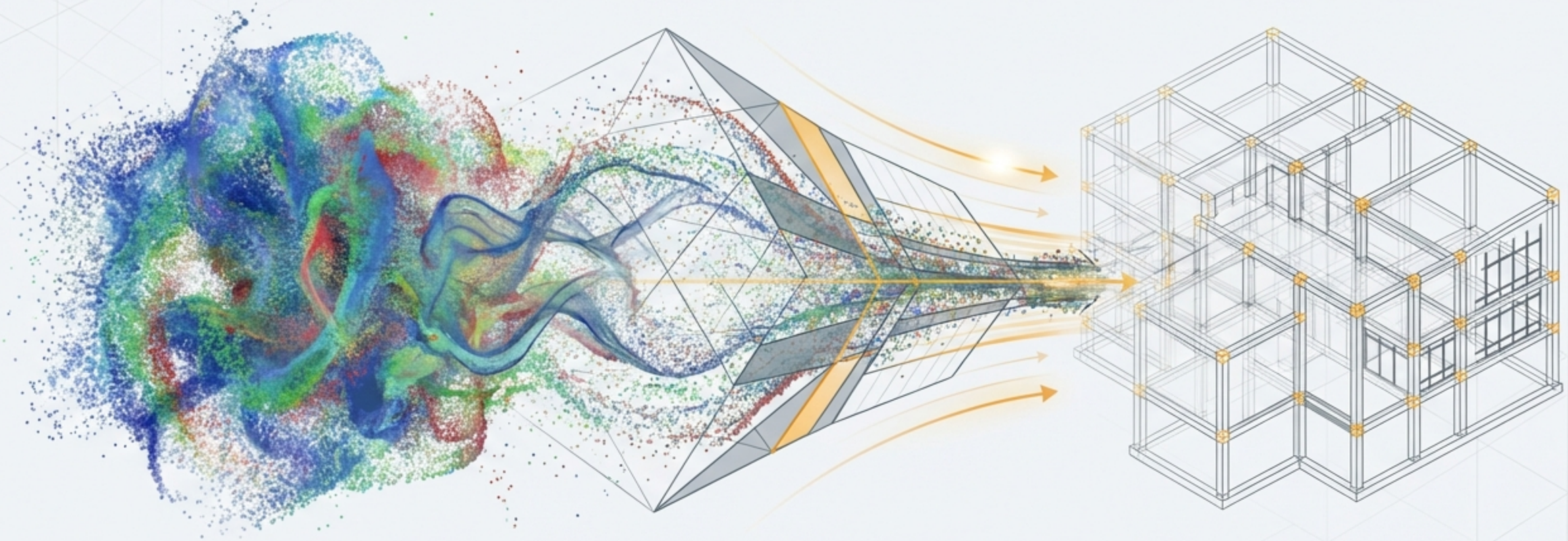
Project GSS: Automated 3D Gaussian Splatting to BIM Pipeline

End-to-End Automated Architecture and Validation for Video-Based Digital Twin Construction



`Python` `Open3D` `gsplat` `IfcOpenShell` `USD`

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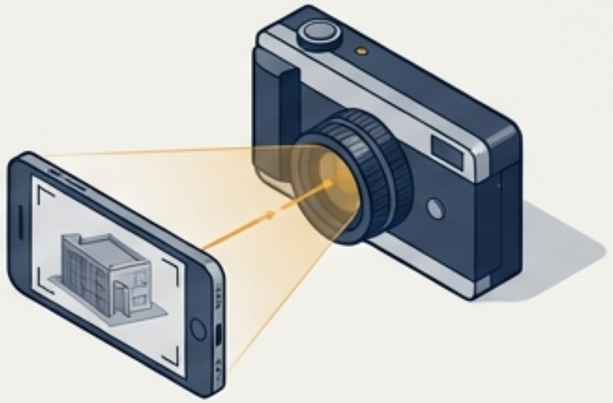
Physical and Economic Limitations of Traditional Scan-to-BIM Pipelines



Traditional Method

- High-cost hardware equipment
- Heavy reliance on manual modeling
- Time-consuming data processing

GSS Solution



- ✓ Camera-based (No LiDAR required)
- ✓ End-to-End automated processing
- ✓ Drastic reduction in time and cost

Key Takeaway: Moving away from expensive hardware and manual workflows, the industry requires a pipeline that automatically generates standard BIM models from a single video input.

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GSS Pipeline Architecture: End-to-End Data Flow

Strictly Typed Pydantic Configs

BaseStep Pattern Architecture



Raw Video Input

Point Cloud Reconstruction

RANSAC-based Detection

Industry Standard BIM

Cross-Platform Export

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Four Pipeline Modes & Strict Step Pattern Architecture



Standard Mode

9-step core flow driven by gsplat (2DGS) for baseline point cloud generation.



PlanarGS Mode

7-step flow utilizing diff-plane-rasterization for heavily optimized surface quality.



Exterior Mode

10-step flow incorporating the s06c module to extract facades and roof structures.



Import Mode

High-speed 5-step flow for directly injecting existing PLY point cloud results.



Architecture Note: All modules strictly inherit from `src/gss/core/step_base.py` and comply with YAML-based I/O contracts ensuring total pipeline robustness.

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Core Technology I: 3DGS Reconstruction & RANSAC Plane Extraction



Scene Representation

PlanarGS / gsplat 2DGS

Example: 21.4M points maximizing planarity for walls, floors, and ceilings.



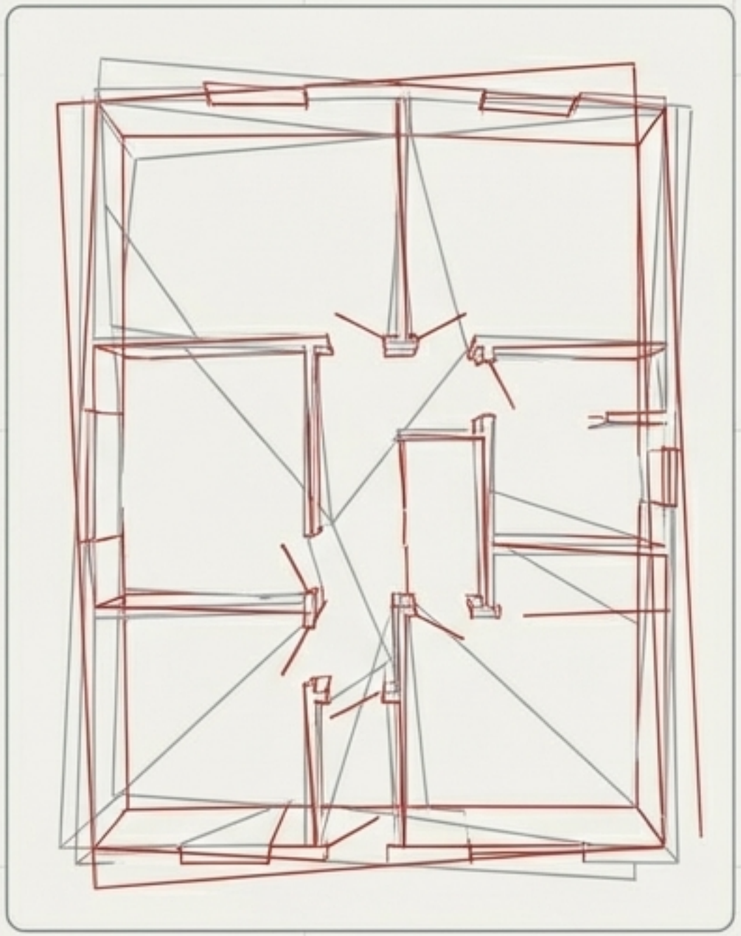
Plane Extraction

Open3D RANSAC + Alpha Shape

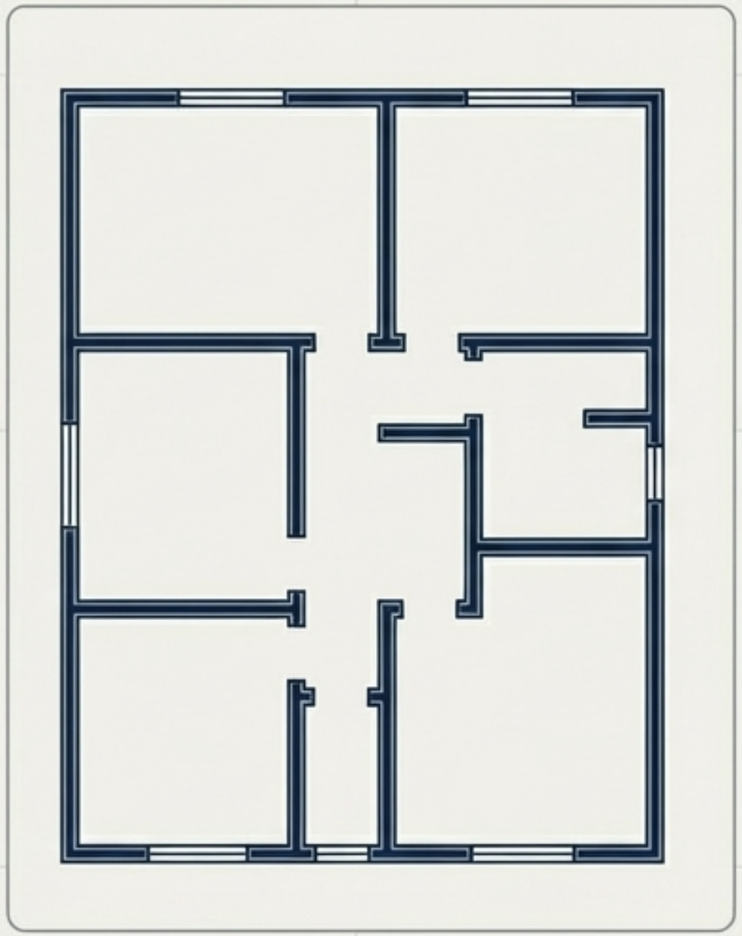
Automatic detection of accurate geometric boundary polygons from noisy point clouds.

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Core Technology II: s06b Manhattan Geometric Regularization Engine



Raw Extracted Geometry



Manhattan Axis Aligned

1. Axis Alignment

Automatic snapping of wall normals to the Manhattan axes (±X/±Z).

2. Height Clustering

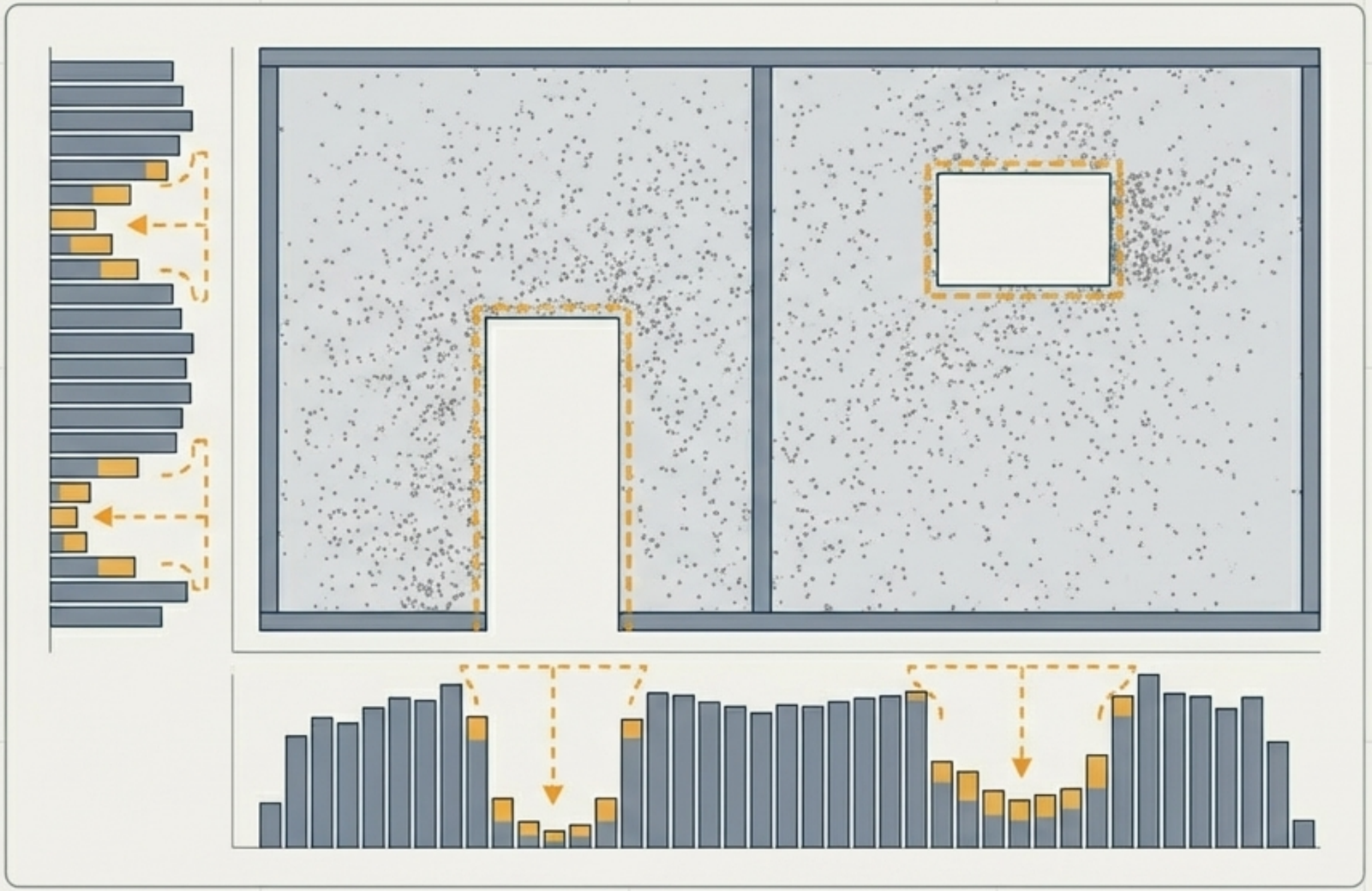
Merging unaligned floor and ceiling heights into consistent Z-axis planes.

3. Thickness & Polylines

Detecting parallel wall pairs to derive center-lines, resulting in perfectly trimmed polygon structures.

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Core Technology III: Space Detection & Cloud2BIM Opening Recognition



Space Detection

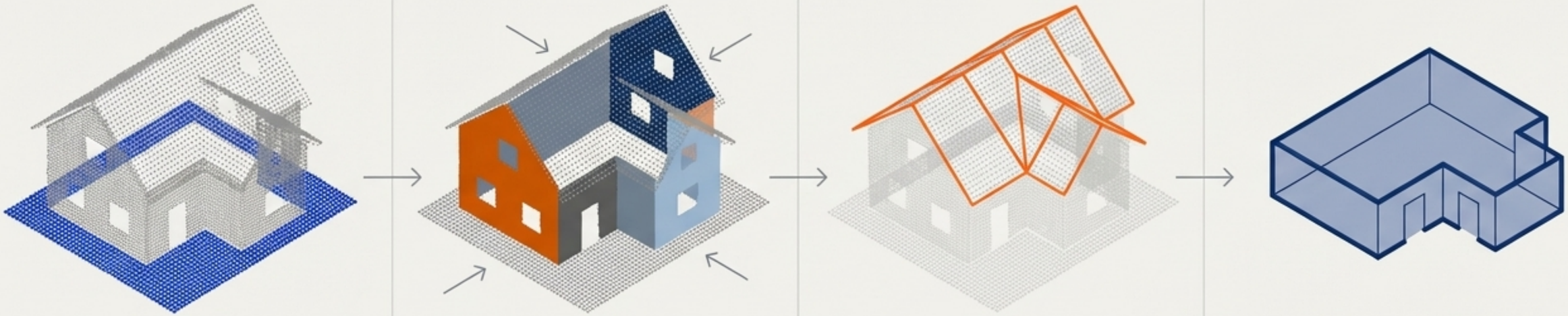
Mathematical exploration and separation of closed room boundaries utilizing **N-point polyline** wall center-lines.

Opening Detection

Analyzing 2D projections and density histograms of surface points to automatically identify and scale doors (height > 1.8m) and windows.

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Core Technology IV: s06c
Exterior Building Structure Extraction



1. Ground Separation
Ground Plane Extraction

2. Facade Clustering

3. Roof Structuring

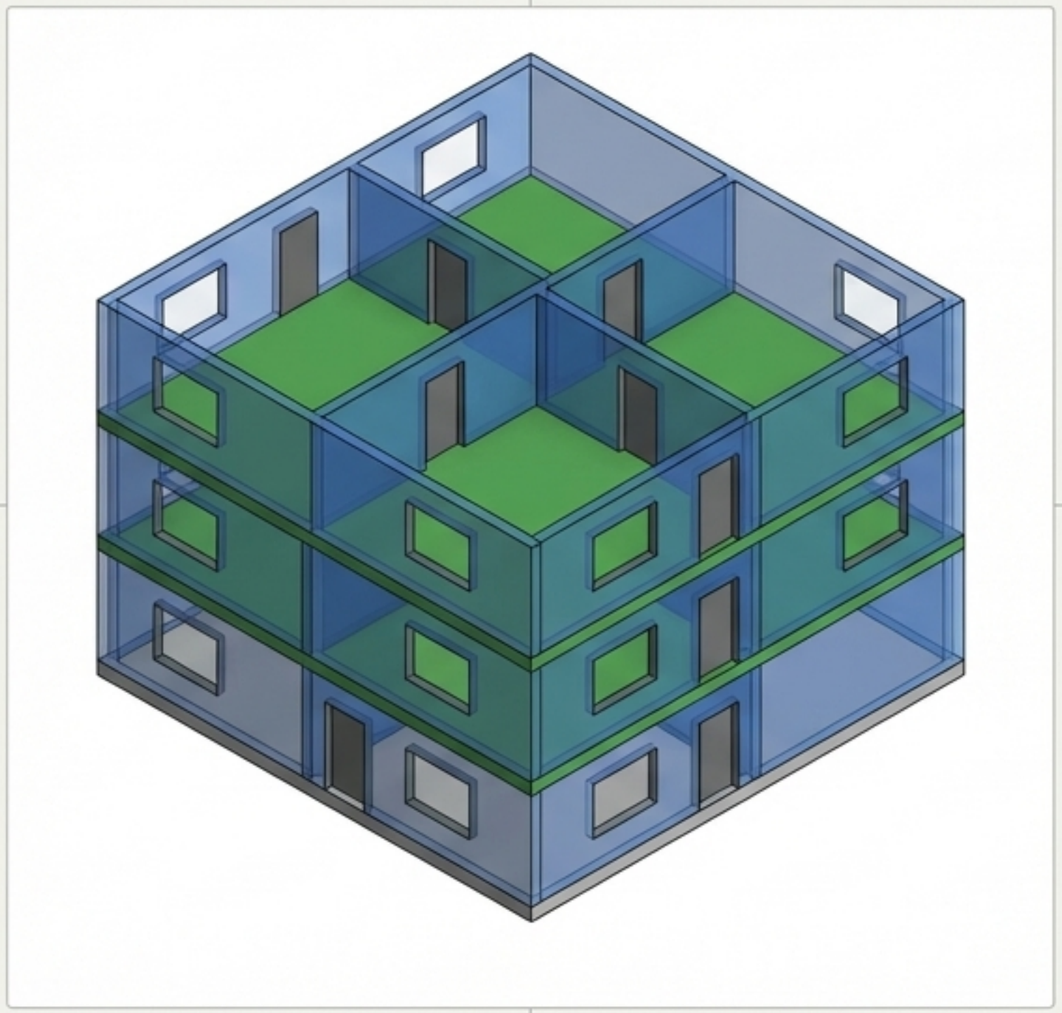
4. Final Footprint

Advanced Mechanics:

- Normal grouping for facade separation
- Concave hull extraction for irregular footprints
- Plane intersection logic for complex roof structures
- Full support for Non-Manhattan oblique geometry

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Core Technology V: Industry Standard BIM via IFC Hierarchy



```
IfcProject
└── IfcSite
    └── IfcBuilding
        ├── IfcWall
        ├── IfcSlab
        └── IfcSpace
```

- Native `IFC4` format generation powered by the `IfcOpenShell` library.
- Uncompromising support for non-orthogonal and multi-segment walls using `IfcArbitraryClosedProfileDef`.
- Precise, Z-coordinate based multi-storey separation mapped directly to `IfcOpeningElement`.

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Core Technology VI: Cross-Platform Compatible Mesh Export

Generated BIM →



GLB

`Powered by trimesh`

Optimized for Web & Game Engines



USDC

`Omniverse Standard`

PBR Materials & Area-Weighted Normals



USDZ

`Apple AR Ready`

Packaged for Spatial Computing

Engineering Specifications:

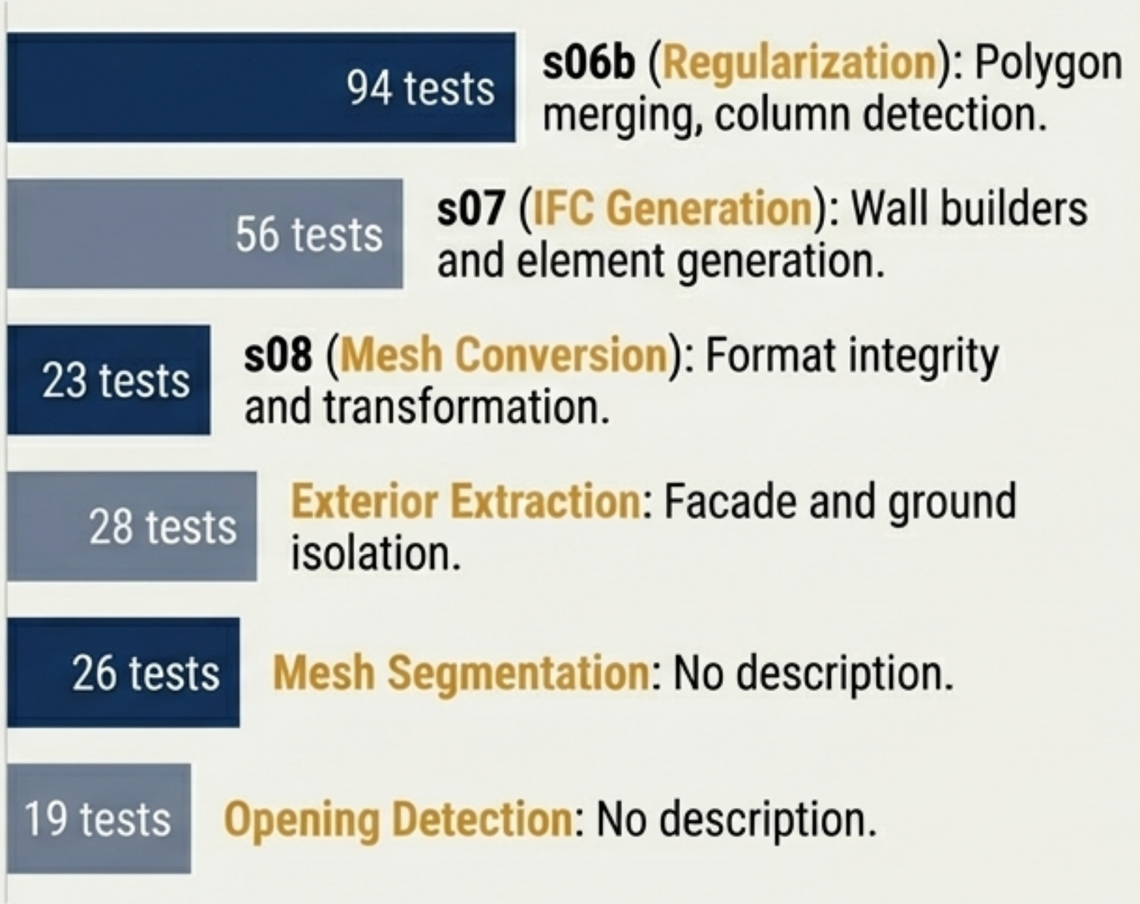
- Mathematical Coordinate Transformation: Executing Z-up to Y-up conversions (x, y, z) → (x, z, -y) to match game engine axes perfectly.
- Strict USD Compliance: Preventative application of `MaterialBindingAPI.Apply()` to ensure zero warnings in Omniverse.
- Graceful Degradation: Built-in fail-safes for `trimesh` and `usd-core` dependencies to ensure pipeline continuity.

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Validation I: Robust Architecture & Comprehensive Test Coverage

365

Total Passed Unit Tests



Proven integrity across the entire end-to-end pipeline.

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Validation II: End-to-End Extraction Across 9 Datasets

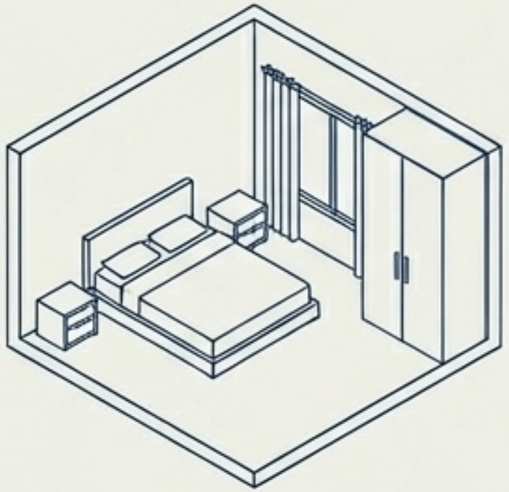
Architectural Datasets

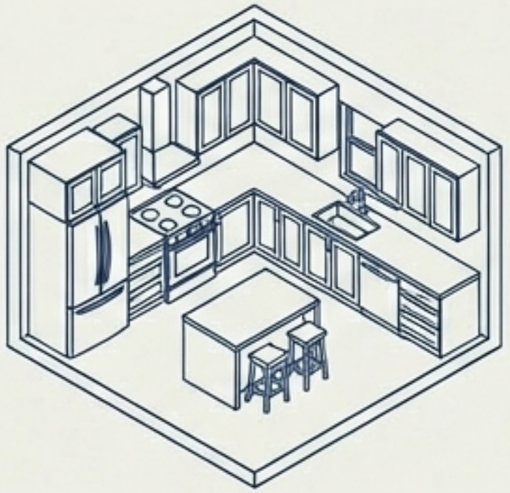
- jh: Perfect extraction (4W, 1F, 1C, 1S)
- drjohnson: Perfect extraction (4W, 1F, 1C, 1S)
- playroom: Perfect extraction (4W, 1F, 1C, 1S)

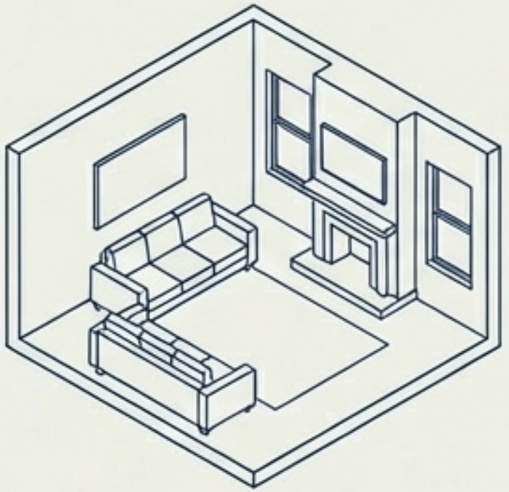
Diverse Environments

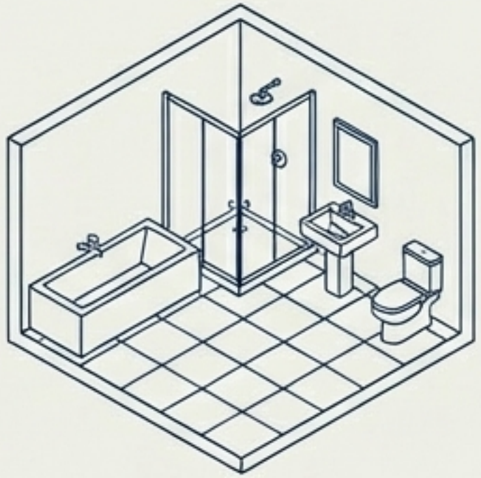
- bonsai
- room
- train
- kitchen
- counter

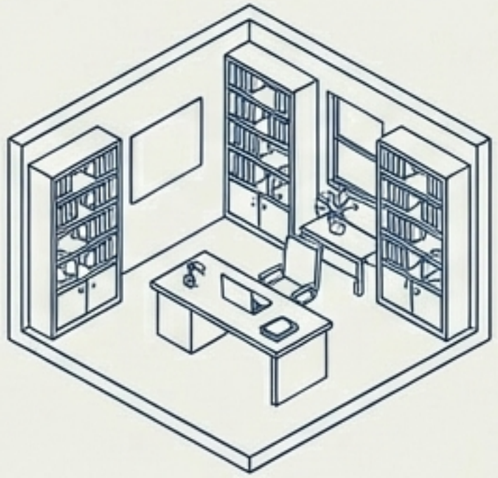
Result: Consistent geometric identification and successful BIM translation achieved across varying scales and environments.

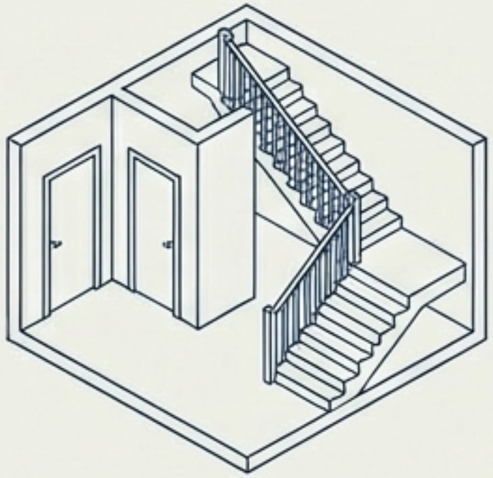








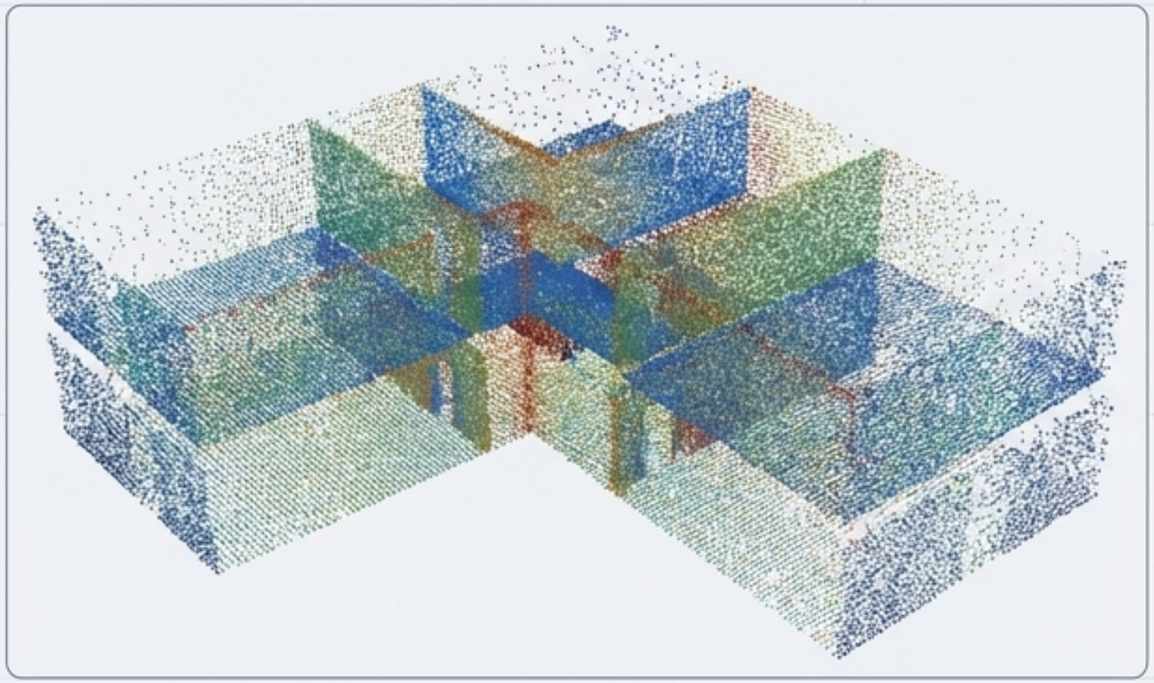




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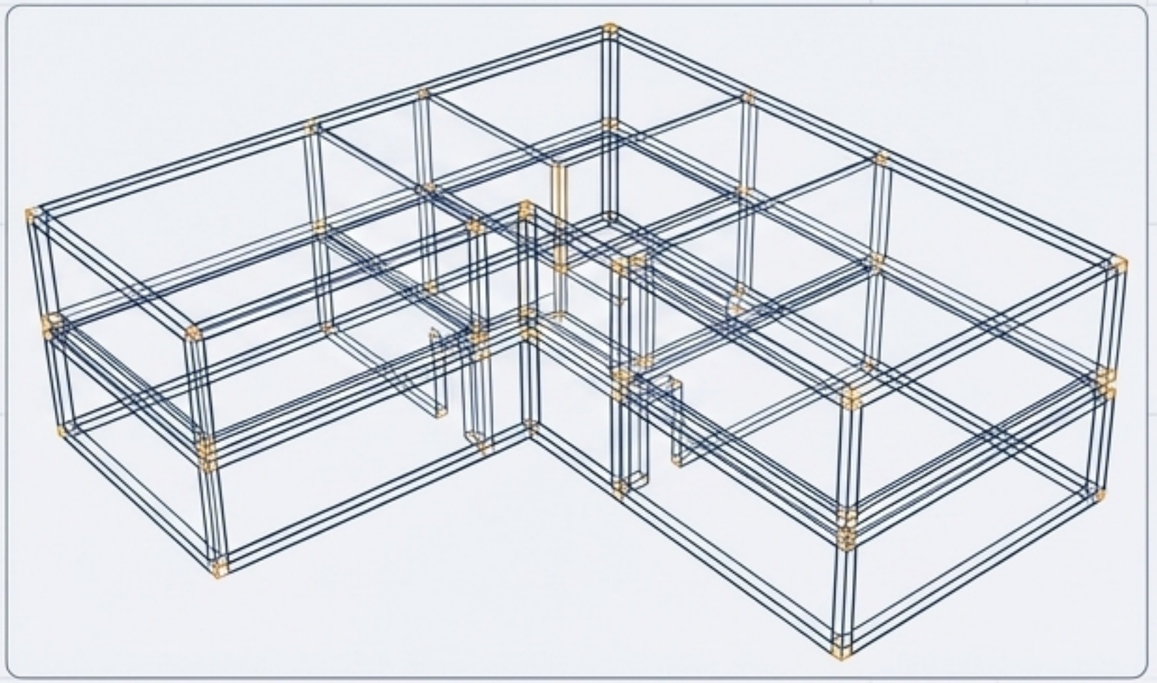
Validation III: Complex N-Point Polygon Structure Verification

Synthetic L-Building Environment



JetBrains Mono: synthetic_L (point cloud)

Automated BIM Output



JetBrains Mono: Generated BIM Model (IFC)

Rigorous Mathematical Verification

- **Test Environment:** Constructed an irregular 'L-shaped' synthetic point cloud (synthetic_L) to stress-test non-rectangular capabilities.
- **Validation Results:** 100% automated detection with zero collisions or omissions.
- **Outputs Generated:** 6 precise wall segments, 3 floor planes, 3 ceiling planes, and 3 distinctly separated spatial volumes.
- **Conclusion:** Proven mathematical capacity to handle complex architectural floor plans far beyond simple rectangular rooms.

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Validation IV: 3DGS ↔ BIM Spatial Coordinate Alignment

Technical Context: Mathematical Alignment Proof

Coordinate Matrix Pipeline:

1. **Native COLMAP coordinates**
2. **Rotation** into Manhattan alignment (pts @ R.T)
3. **Applied IFC Transformation Matrix** (mx/s, mz/s, my/s)

Overlay Test Results:

Using the 'jh' dataset, an **80,000 point** overlay test confirms that the Gaussian point cloud cloud resides flawlessly within the exact boundaries of the extracted BIM volume. This provides **undeniable visual and mathematical proof** of absolute coordinate alignment.



JetBrains Mono: jh (3DGS ↔ BIM Overlay)

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Target Platforms: Immediate Spatial Computing & Digital Twin Integration

GSS Pipeline Output

NVIDIA Omniverse / Isaac Sim



Native **USDC (Y-up)** integration designed specifically for physics simulation and robotics environment construction.

USDC (Y-up)

JetBrains Mono

Apple Vision Pro



Packaged **USDZ** export for immediate, high-fidelity Spatial Computing and Augmented Reality visualization.

USDZ

unity

Universal **GLB** format compatibility for game engines and lightweight web-based Digital Twin viewers.

GLB

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Future Directions: The Next Frontier for Physics-Ready Digital Twins

Semantic Segmentation Advancement

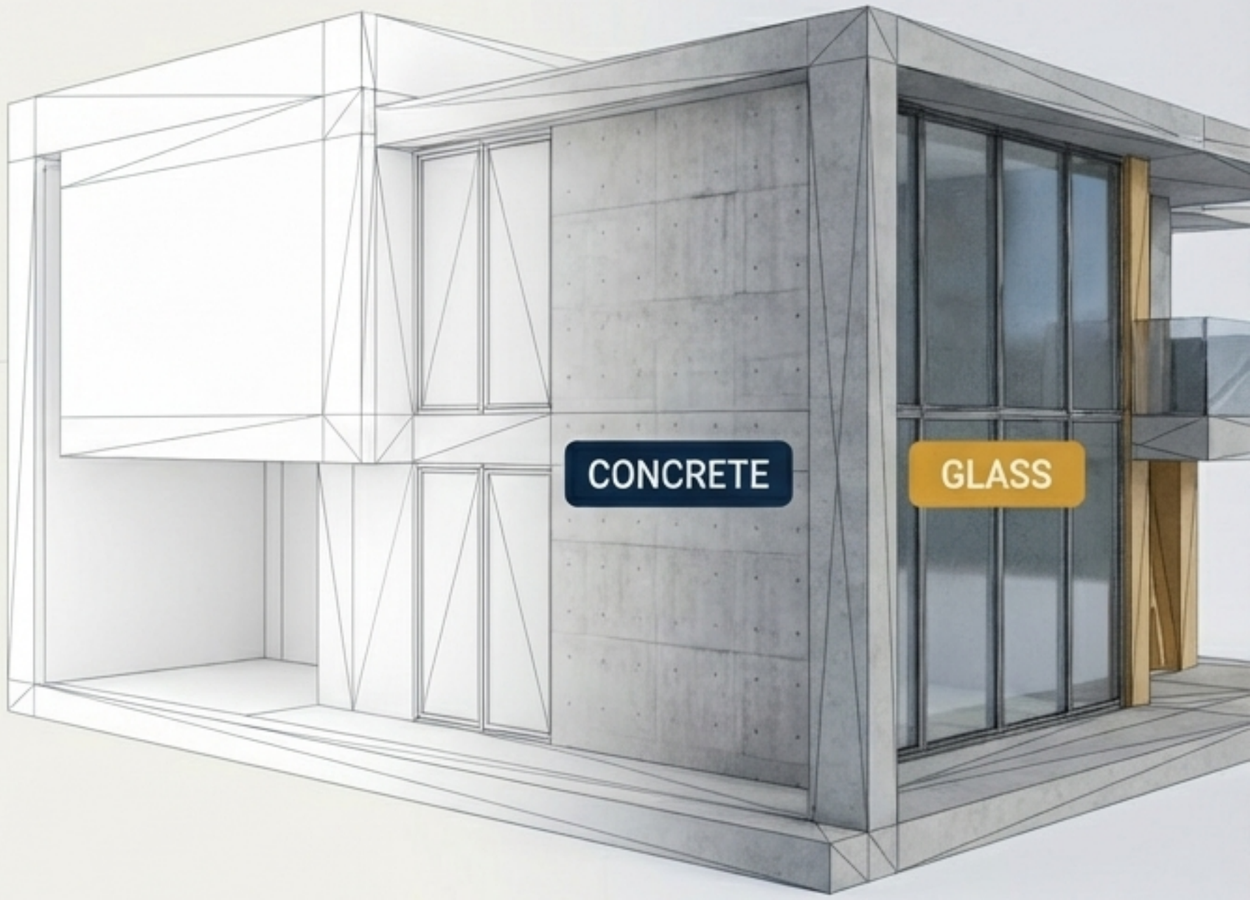
Evolving beyond basic geometric opening detection to enable AI-driven, high-precision identification of doors and windows at LOD 300+.

Material Classification Integration

Deploying Vision-Language models (SAM2 + CLIP) to automatically classify and map surface materials such as concrete, glass, and wood.

The Ultimate Goal

Moving past pure geometry to complete a ‘Physics-Ready’ 3D data pipeline that fundamentally encodes material and physical properties for advanced simulation.



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